

Mean



1 It measures the center/average of the scores

2

a 11.8

b 77.25

c 19.72

d -7.25

3 56

4 Greater

5 Smaller

6 19.87

7

a 1093.3

b 40

8

a 16

b 18

9

a

b 8.91

Score (x)	Frequency (f)	fx
5	14	70
7	4	28
9	2	18
11	18	198
13	6	78
Totals	44	392

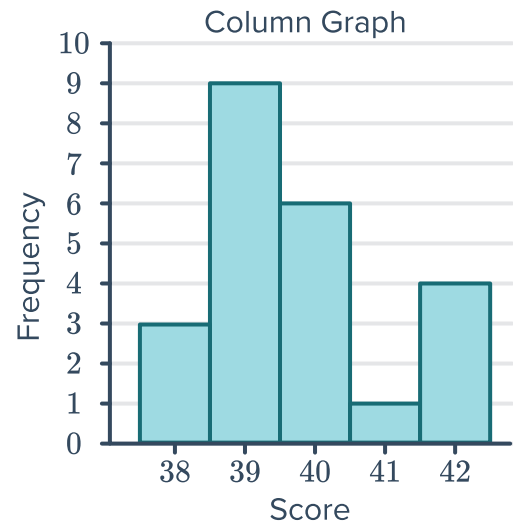
10

a

Score (x)	Frequency (f)	fx
38	3	114
39	9	351
40	6	240
41	1	41
42	4	168
Total	23	914



b



c 39.7

Median

11 11

12

a 35

b Between 76 and 77

13 $n = 48$

14

a

i 46.6, 52.8, 63.9, 66.7, 69.4, 76.6, 80.6

ii 66.7

b

i $-77.6, -54.8, -42.6, 40.6, 44.9, 45.6, 67.9, 74.7$

ii 42.75

c

i 1, 3, 11, 16, 16, 24, 24, 28

ii 16

15 37, 39, 41, 43

16

a 34

b 48

17



a

i 67

ii 45

iii 3062

iv 67

b

i 60

ii 4.5

iii 236

iv 3.9

18 83.5

19

a 25

b 18

c 13

20

a No

b No

c Yes

d No

Mode

21

a 8, 14

b 26

c 3.0

d 71

22

a 74.07 hours

b 73 hours

c 67 hours

Measures of center

23

a Mean, because there are no outliers.

b Mean, because there are no outliers.

c Mean, because there are no outliers.

d Median, because there is an outlier.

24

a Median = 657

b Mean = 108.8

25 \$1.64

26

a Mode = Median = 11

b Mean = Median = Mode = 90

c Median = 22

Outliers



27 The mode will not change.

28 The outlier was smaller than the remaining values in the data set.

29

a \$527 000

b \$929 000

c \$446 600

Applications

30

a

Heart Rate	Class center (x)	Frequency (f)	$f \times x$
30 – 39	34.5	13	448.5
40 – 49	44.5	22	979
50 – 59	54.5	24	1308
60 – 69	64.5	36	2322

b 53.24

31

a 86

b 84.8

c 85.4

d 97, Student A

e 72, Student B

32

a \$399 400

b 40%

c \$386 500

d 50%

33 $x = 78$

34 63.65

35

a 15

b 13.29

c Median, because there is an outlier.

36

a 5 employees

b 58

c 20

d 36.5

e 20%



37

- a 90 minutes b 25% c \$80 000

38

a

Weight in kilograms (x)	Frequency (f)	fx
16	12	192
17	17	289
18	21	378
19	10	190
20	18	360
21	22	462
Total	100	1871

- b 18.5 kilograms
c 18.71 kilograms
d Mean

39

- a 19 b 14 minutes c 3 minutes d 7.42 minutes
e 5 minutes f 7 minutes